

2026 AUSTRALIAN JUNIOR CHESS CHAMPIONSHIP PROBLEM SOLVING ALPHA CANBERRA

NAME: _____

DATE OF BIRTH: _____

Please circle your age group:

- U8 (born in 2019 or after) U14 (born in 2013 or after)
- U10 (born in 2017 or after) U16 (born in 2011 or after)
- U12 (born in 2015 or after) U18 (born in 2009 or after)

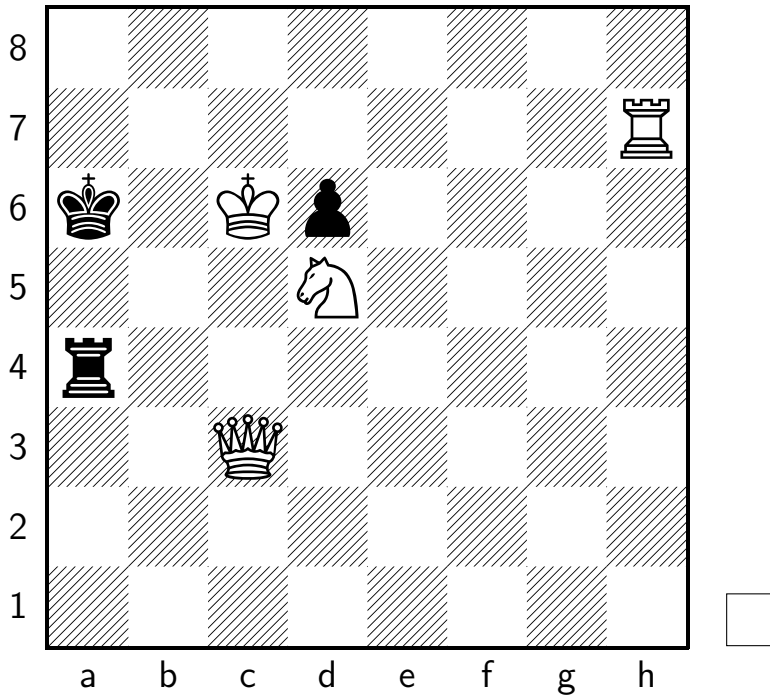
Please circle: Boy or Girl

Please read the following rules.

- You have up to 2 hours for the problem solving paper. You may not need all that time. You may leave after handing in your answers.
- The questions generally go from easy to hard, so it's best to start from the beginning and go in order.
- You can use your chess board and pieces to help you.
- Each page show a question and the marks you can score. Please write your answers in the space under each diagram. Altogether, the test has 100 marks. Even if you don't give perfect answers, you can still get some partial marks.
- For endgame puzzles, show enough moves to reach a position where you are clearly winning or drawing.
- You don't have to finish them all, do your best and solve as many puzzles as you can.

Enjoy the problem solving.

Q1. Mate in 2. (4 marks)

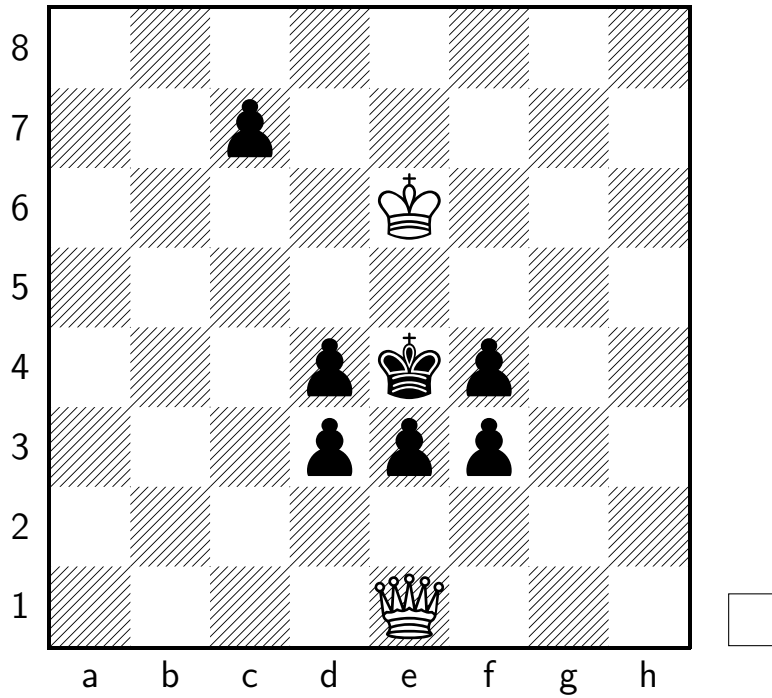


Answer:

A variation 1 ♔d3+ ♖c4 2 ♔xc4+ ♙a5 3 ♔b5# takes 3 moves. The right answer starts with a queen sacrifice, 1 ♔a5+ ♖xa5 2 ♘b4#. If Black king takes the White queen, 1... ♔xa5 2 ♖a7#.

It is very important that you can only get full marks when you find the checkmate within the required moves (in this question, 2 moves, not 3 moves.)

Q2. Mate in 2 (4 marks)



Answer:

1 ♔a5! covers all Black's moves.

1... d2 2 ♔f5#

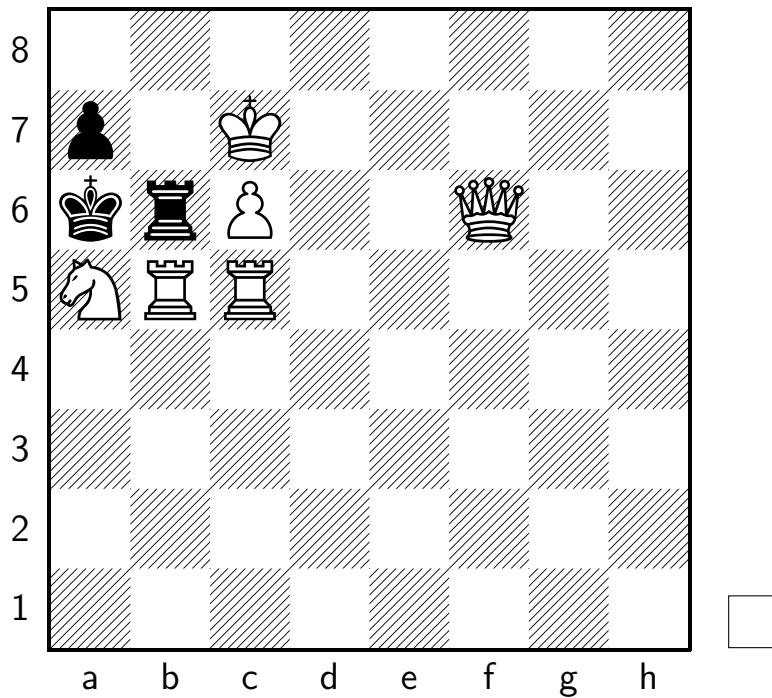
1... e2 2 ♔e5#

1... f2 2 ♔d5#

1... c5 2 ♔a8#

Source: Thomas Salthouse, Falkirk Herald, 1914 (YACPDB: 22264)

Q3. Mate in 2. (6 marks)



Answer:

1 ♔f1

1... ♖xb5 2 ♔xb5#

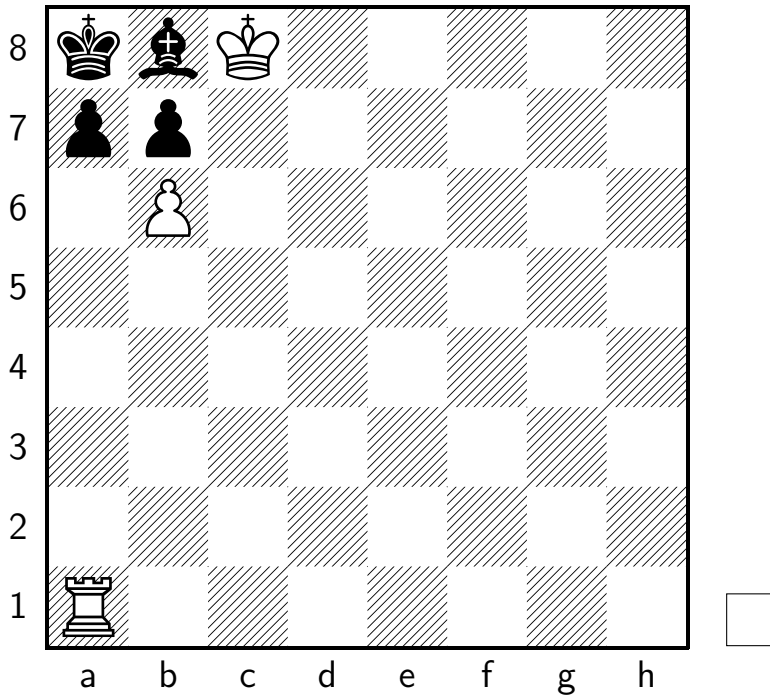
1... ♖xc6+ 2 ♖xc6#

1... ♖b7+ 2 ♖xb7#

1 ♔a1 fails to 1... ♖xb5 2 ♘b3+ ♖a5 3 ♖xa5# (3 ♔xa5#)

Source: Cornelis Goldschmeding, problem (Zagreb), 1957 (YACPDB: 8721)

Q4. Mate in 2. (6 marks)

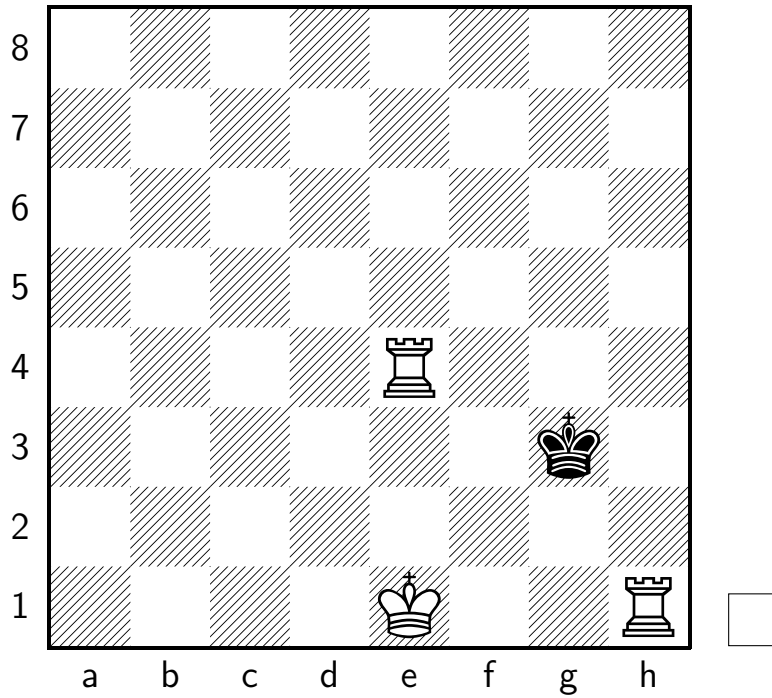


Answer:

This is one of the most famous and hard to solve "Mate in 2" puzzles. The surprising first move is **1 ♖a6!**. If the b7 pawn takes the rook, **1... bxa6 2 b7#**. If the bishop moves, **1... ♗h2 2 ♖xa7#**. The bishop moves to any legal squares (c7, d6, e5, f4, g3) are also correct answers.

Source: Paul Morphy, 1910 (YACPDB: 625426)

Q5. Mate in 2. (6 marks)



Answer:

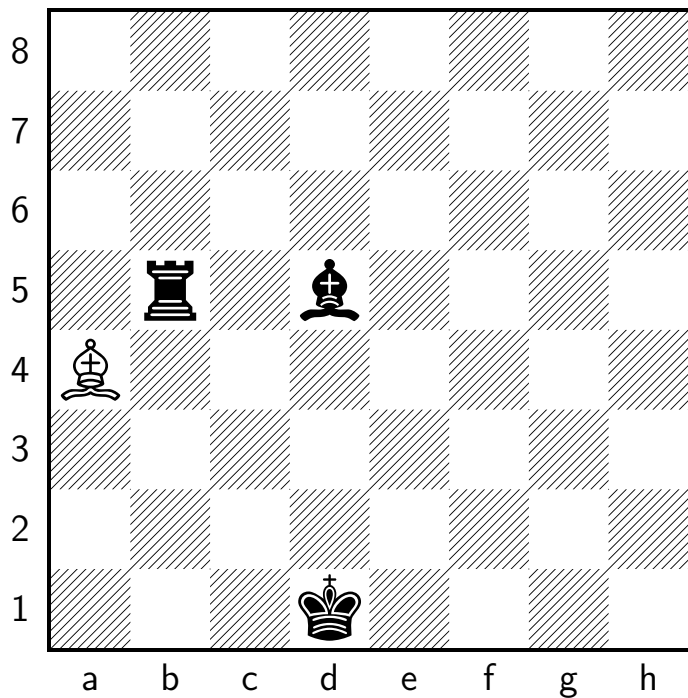
1 O-O ♔h3

2 ♖f3#

Please remember that, unless otherwise stated, castling and en passant should be considered in chess problems.

Source: W. E. Candy, 1911 (YACPDDB: 3889)

Q6. What is the missing piece? Whose turn is it? Which square does the missing piece belong on? Show the previous moves that reached this position. (6 marks)



Answer:

Source: Raymond M. Smullyan: The Chess Mysteries of the Arabian Knights

Q6. Answer:

Article 1.2 of "Laws of Chess" says "Leaving one's own king under attack, exposing one's own king to attack and also 'capturing' the opponent's king are not allowed."

1. The King can not be captured, so the one missing piece is the White king.
2. The Black king is in check by the White bishop. You can not leave your own king under attack. Therefore, it is Black's move now.
3. What was the previous White's move? It is not the White's bishop, so the White king moved from b3, which creates a discovered check to the Black king.
4. The White king on b3 is under double attack. The only possible sequence is for White to play c2-c4 to block the bishop's check, then Black plays bxc3 en passant. White then captures the Black pawn on c3 with the king.

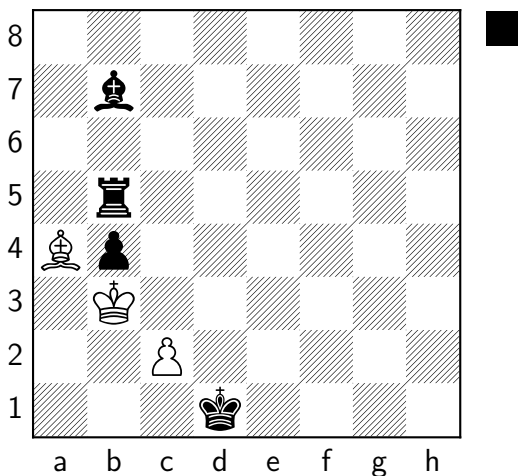
-1. Kb3-Kxc3+

-2. bxc3 e.p.

-3. c2-c4

-4. *-Bd5+ (* means any legal positions)

Below is the initial position and forward moves.

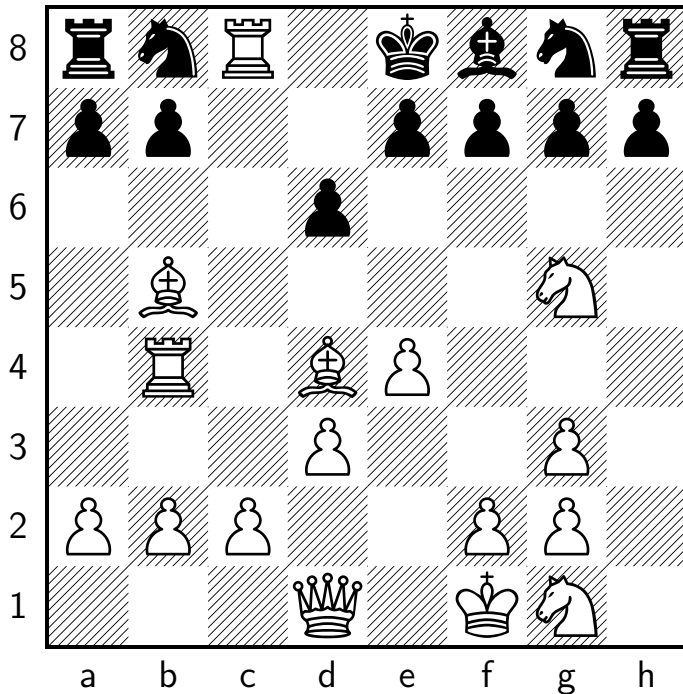


1... ♖d5+

2 c4 bxc3+

3 ♔xc3+

Q7. WHO moved last? WHICH piece was moved from WHERE to WHERE? (6 marks)

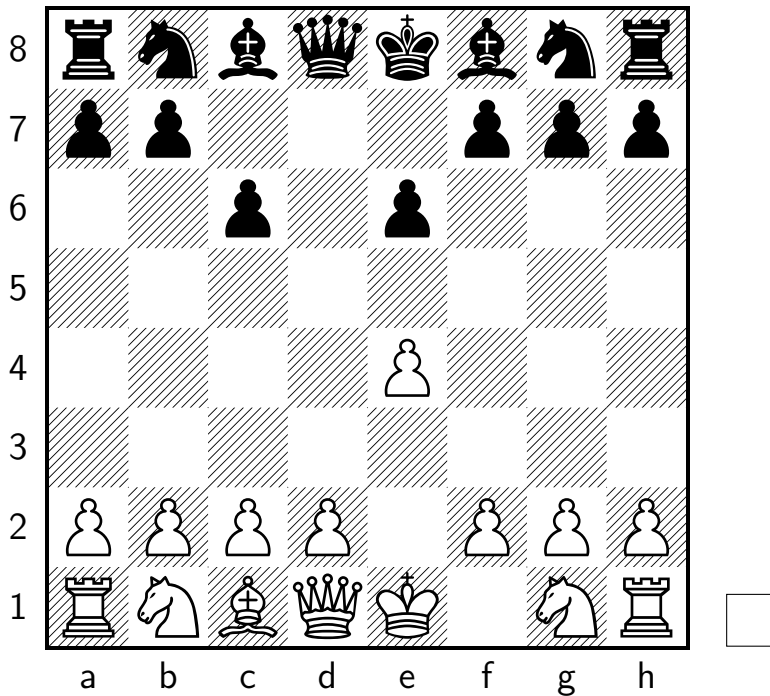


Answer:

The Black king is now in check, and is actually under double check from Rc8 and Bb5. It is illegal to move into check or leave your king in check. Therefore, the last move is made by White. Double check always comes from a discovered check and a check by that move. The only possible move is a move of the rook from c6 to c8.

Source: Noam Elkies, U.S. Problem Bulletin, 1995

Q8. The position below occurred after the Black's 4th move, from the starting position. Show the moves to reach this position. (6 marks)



Answer:

This type of chess problem is called "Proof game", a type of retrograde analysis. This position is easily reached by 1 e4 e6 2 ♖b5 c6 3 ♙xc6 dxc6. However, this problem is asking for the same position after Black's 4th move. There should be illogical, but legal White and Black moves. The answer is

1 e4 e6.

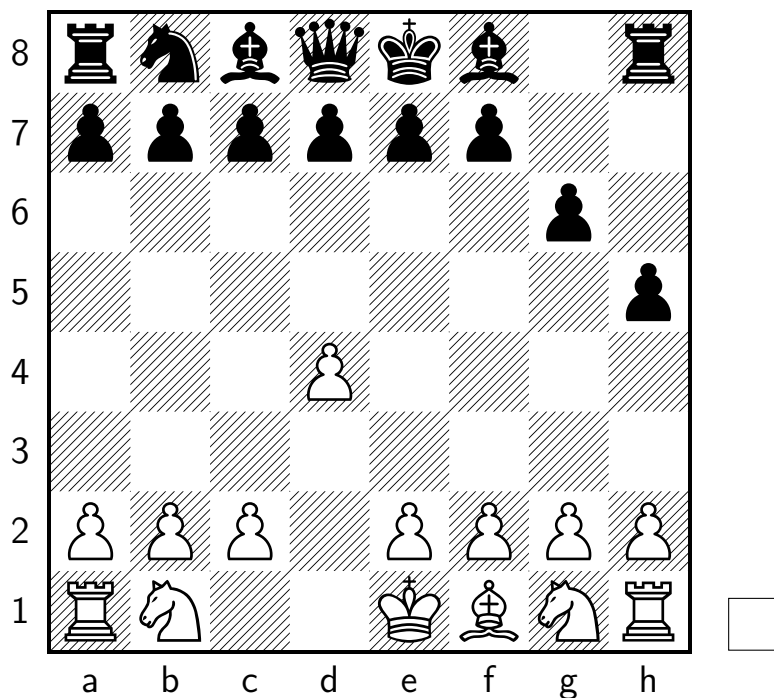
2 ♙b5 ♔e7.

3 ♙xd7 c6.

4 ♙e8 ♔xe8.

Source: Tibor Orbán, Die Schwalbe, 1976

Q9. The position below occurred after the Black's 4th move, from the starting position. Show the moves to reach this position. (6 marks)

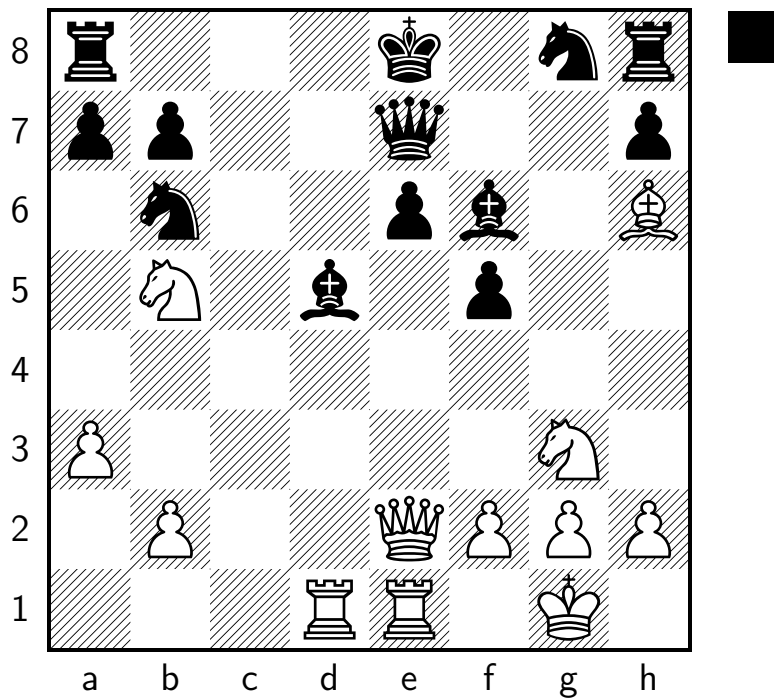


Answer:

- 1 d4 ♘h6
- 2 ♙xh6 g×h6
- 3 ♔d3 h5
- 4 ♔g6 h×g6

Source: L. Mano, source unknown, 1999

Q10. Helpmate in 1. What is Black's move to allow White a "Mate in 1"? Answer Black move's first, then White's checkmates. (6 marks)

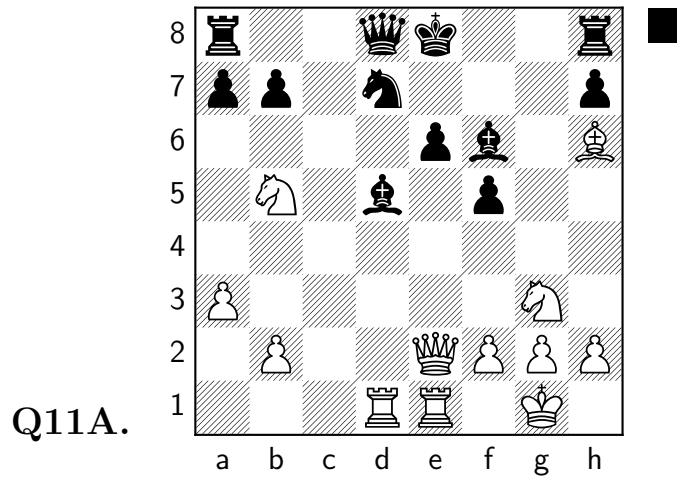


Answer:

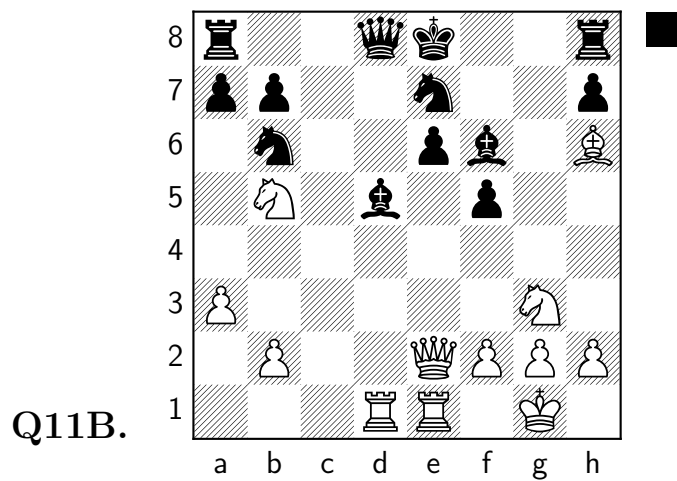
1... ♔f7!

2 ♚h5#

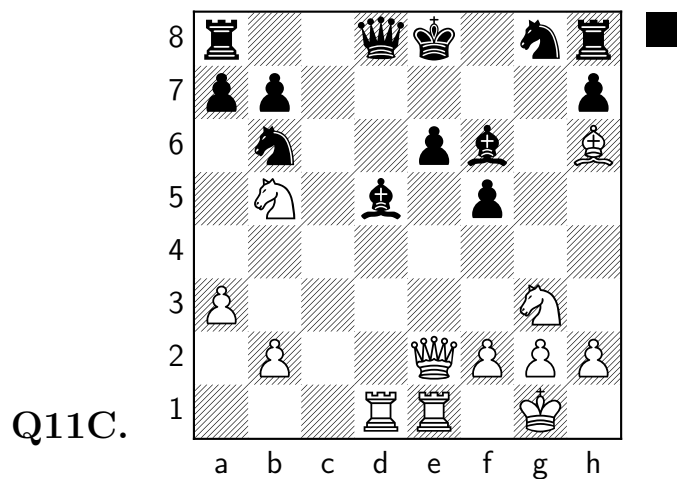
11. Helpmate in 1. Black's positions are all different, but White's positions are all same as the previous question. (2 marks each)



Answer: 1... ♖e7! 2 ♔h5#



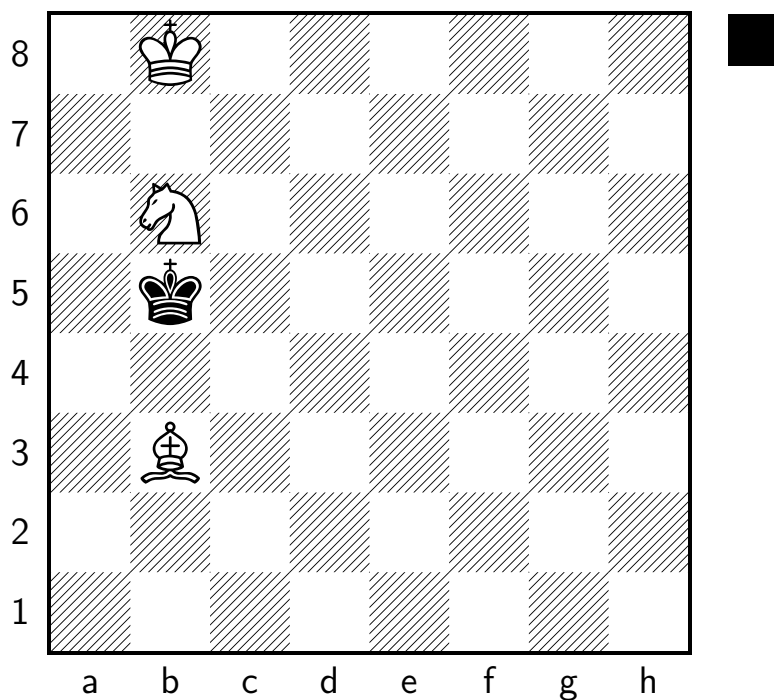
Answer: 1... ♘d7! 2 ♘d6#



Answer: 1... ♔d7! 2 ♚xe6#

Source: Jeff Coakley, Winning Chess Puzzles for Kids Vol. 2

Q12. Helpmate in 3. Black moves 1st and White checkmates with their 3rd move. (6 marks)



Answer:

0... ♔a6?

1 ♔c7 ♔a7?

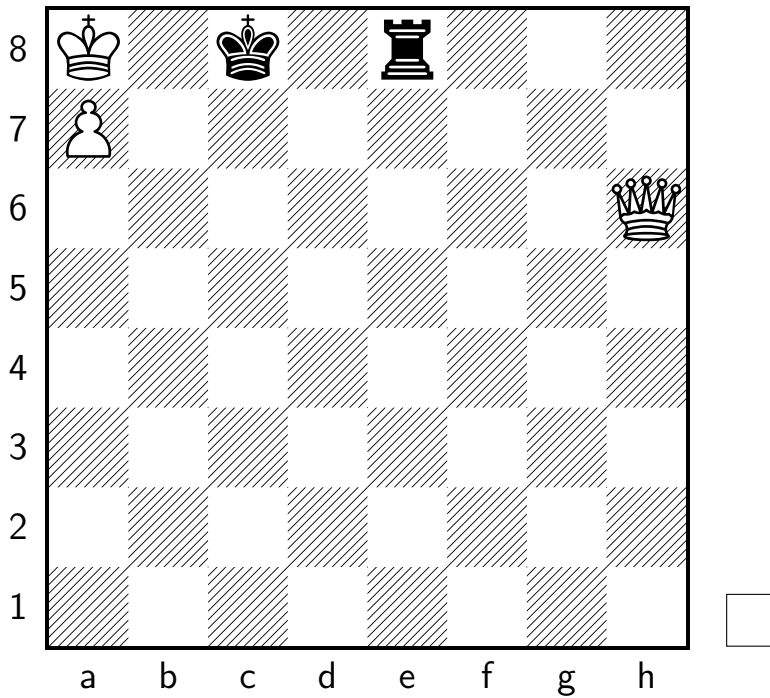
2 ♘c8+ ♔a8??

3 ♚d5#

With Black's help, knight and bishop checkmates are quite easy!.

Source: Hilmar Ebert, Schach-Echo, 1977 (YACPDB: 638060)

Q13. Selfmate in 2. White forces Black to checkmate White's King with Black's 2nd move. (NB This must be a forced variation as Black will try and avoid checkmating White if possible!) (6 marks)

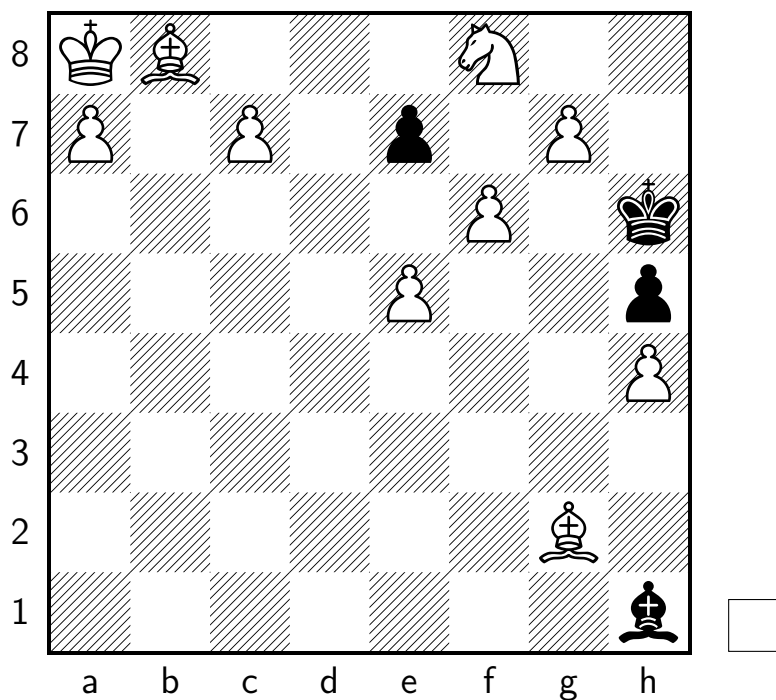


Answer:

- 1 ♔c6+ ♚d8
 2 ♔c7+ ♚xc7#

Source: Wilhelm Krämer, Die Schwalbe, 1928 (YACPDB: 106430)

Q14. Selfmate in 2. White forces Black to checkmate White in 2 moves. (6 marks)



Answer:

1 c8♖ exf6

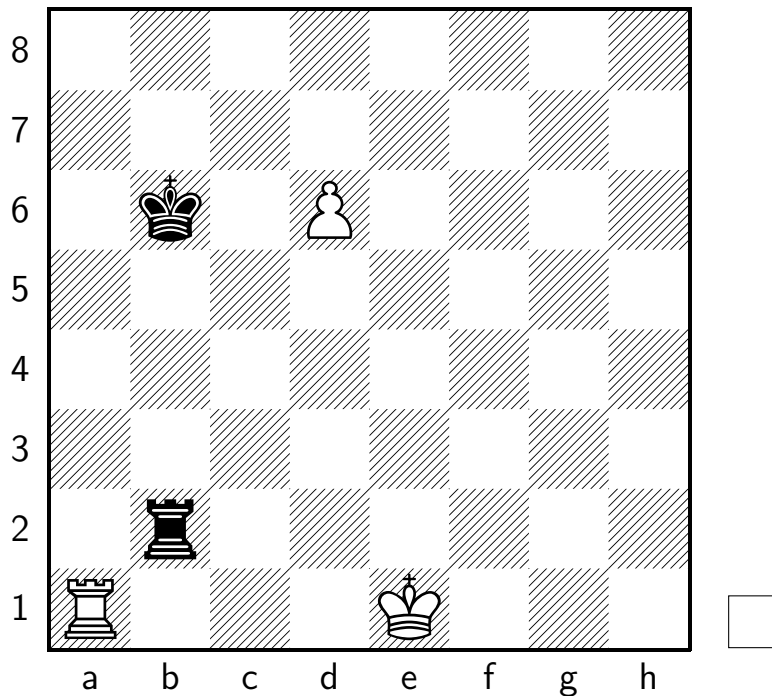
2 exf6 ♜xg2♯

If 1... e6, another waiting move is **2 g8♜**, then **2... ♜xg2♯**.

Do you understand why two underpromotions are necessary?

Source: Wolfgang Pauly, The Theory of Pawn Promotion, 1912 (YACPDB: 47304)

Q15. White play and win (6 marks)



Answer:

Castling first does not work.

1 O-O-O ♖a2

2 d7 ♖a1+

3 ♔c2 ♖xd1

4 ♔xd1 ♔c7

The answer is as below.

1 d7 ♔c7

2 d8 ♕+ ♔xd8

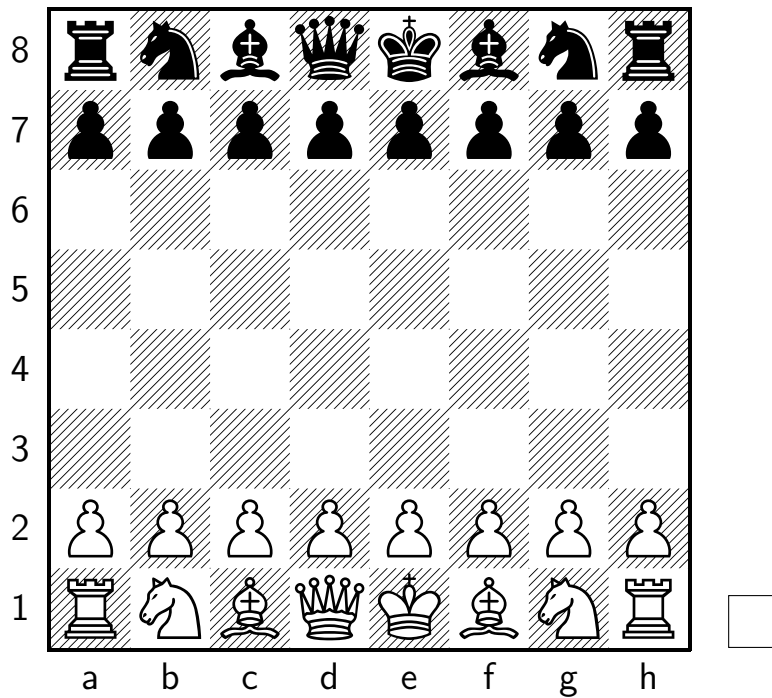
3 O-O-O+ ♔e7

4 ♔xb2 (White is winning with a rook.)

For this type of problem or study, you do not have to show all the moves up to checkmate. However, please write down enough moves to show a clear winning or drawing position.

Source: Alexey Selezniev, Tidskrift for Schack, 1921 (YACPDB: 384775)

Q16. White to play. Who wins?



Answer:

This is the world's hardest chess problem. We hope you will be one step closer to the answer during the 2026 Australian Junior Championships.

Happy Problem Solving!

Source: Composer unknown, date unknown